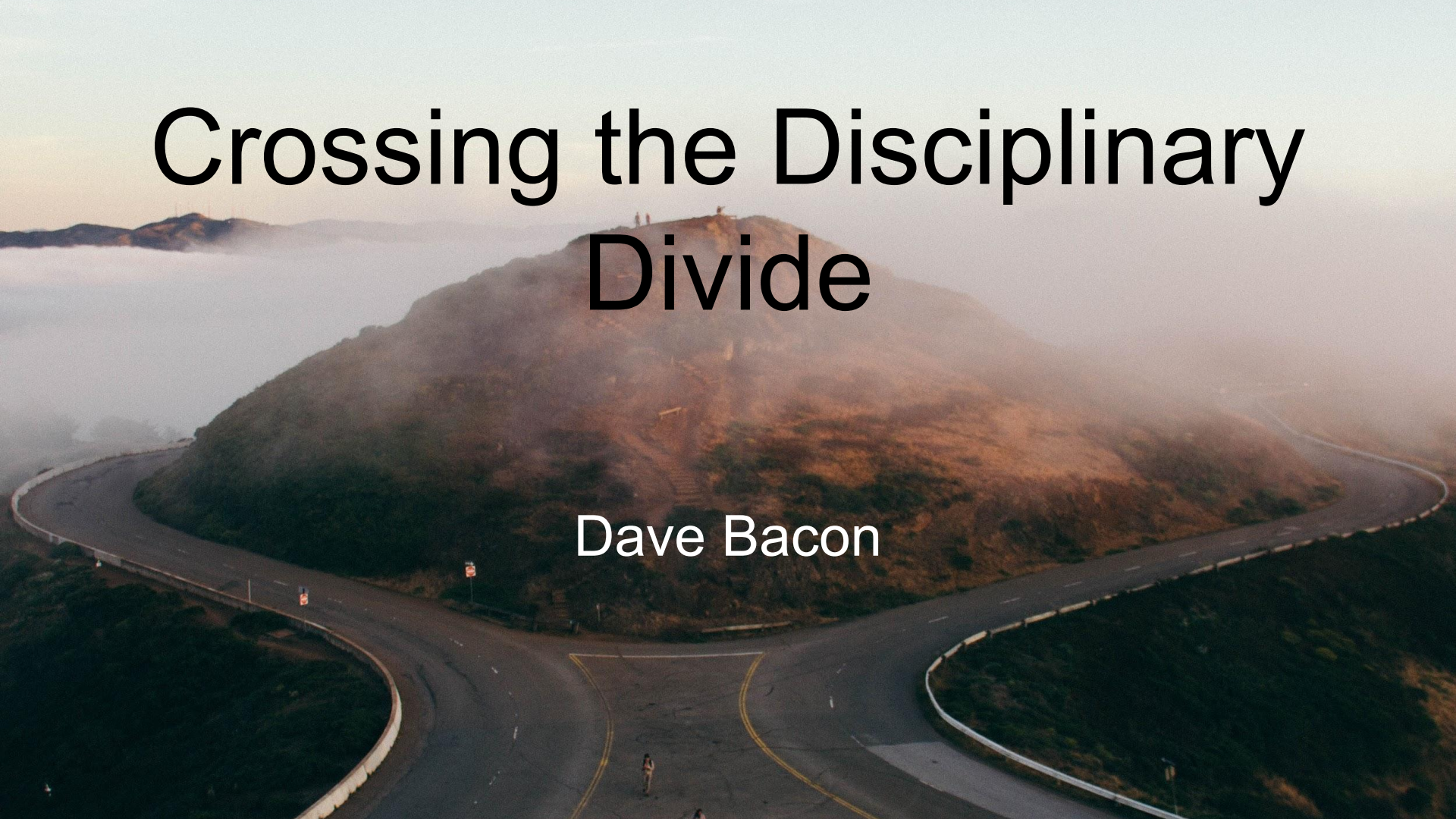
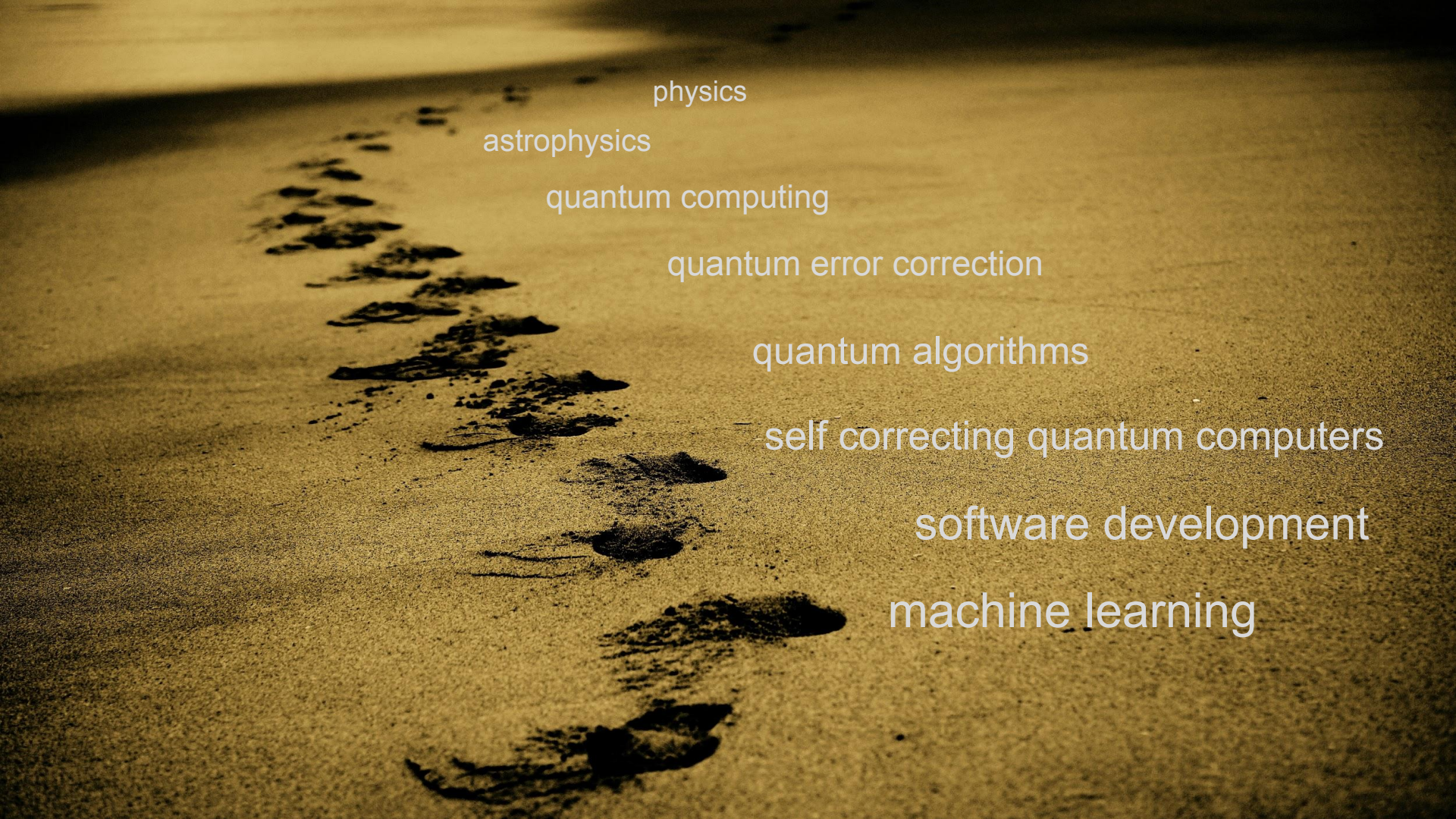


Crossing the Disciplinary Divide

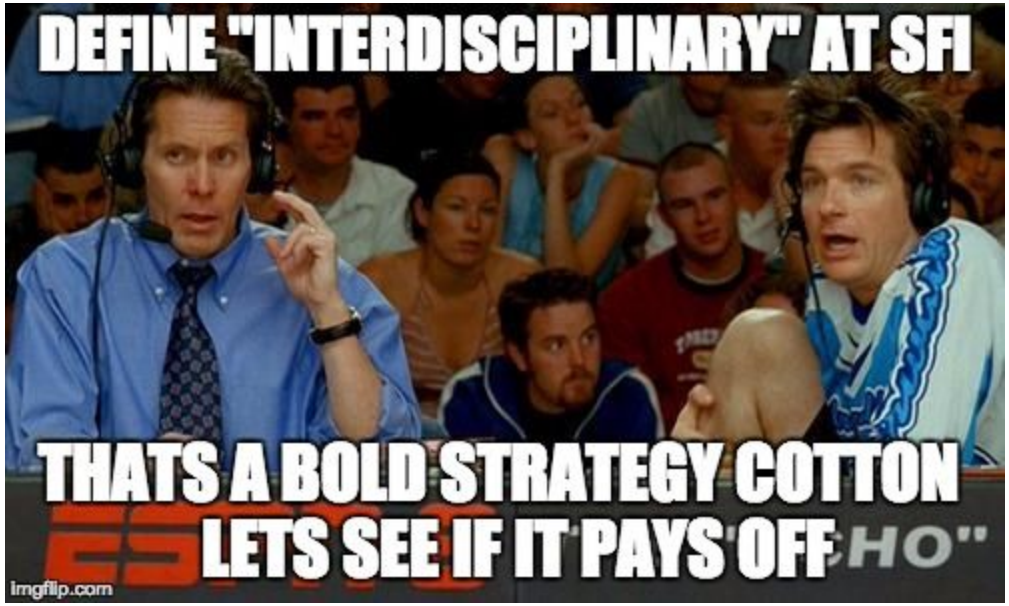
Dave Bacon

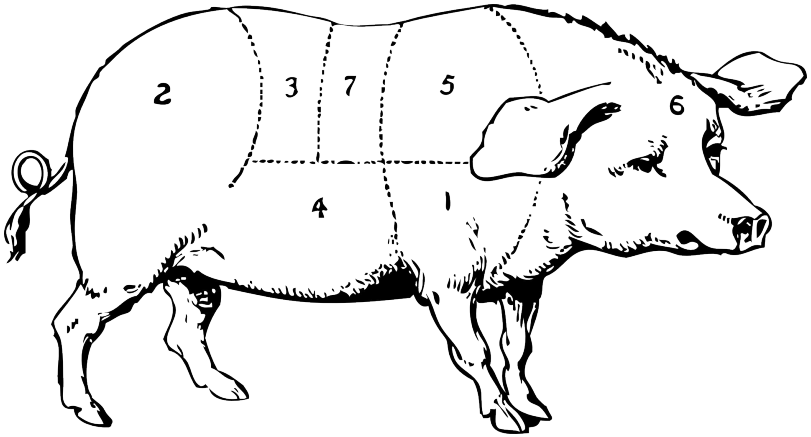

B.S. Physics and Literature, Caltech (93-97)
Ph.D. Physics, Berkeley (97-01)
Postdoc, Institute for Quantum Information, Caltech (01-04)
Postdoc, Sanfe Fe Institute (04-05)
Scientist, CS, University of Washington (05)
Assistant Research Prof, CS (06-11)
Assistant Research Prof, Physics (07-11)
Software Engineer, Google (11-)


physics
astrophysics
quantum computing
quantum error correction
quantum algorithms
self correcting quantum computers
software development
machine learning

What I Know Now That I Wish I Knew As A Postdoc

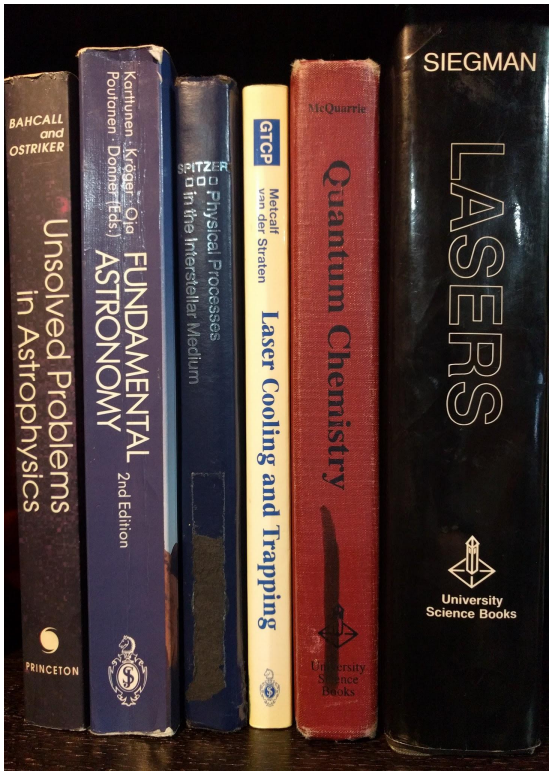
- A framework for cross disciplinary work in academia
  (and how it impacts getting a job, and how you work)
- Investing in a dictionary
- Freedom, hobbies, and of endeavors of a whimsical nature
- Create, create, create, create


DEFINE "INTERDISCIPLINARY" AT SFI
THATS A BOLD STRATEGY COTTON
LETS SEE IF IT PAYS OFF
imgflip.com



Multidisciplinary	join to solve problems within a discipline
Interdisciplinary	join to form new problems and disciplines
Transdisciplinary	use of other discipline to undermine a discipline

Multidisciplinary



trapping of single molecular ions

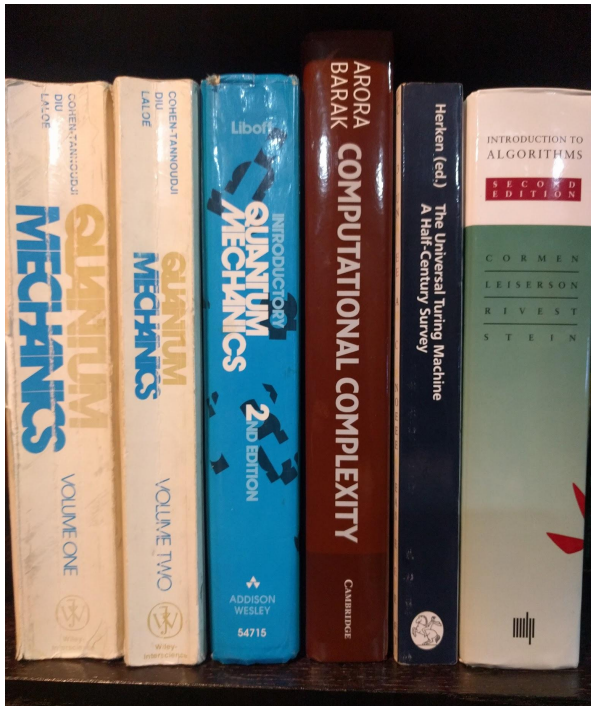
+

molecular ions as probe of physics of the interstellar medium

=

better theoretical models of interstellar medium

Interdisciplinary



quantum theory of physics

+

theory of computational complexity

=

quantum complexity theory

Transdisciplinary



topological quantum field theory

+

theory of knot invariants

=

new basis for theory of knots

Phthi·sis /ˈTHīsəs,ˈtīsəs/

Syz·y·gy /ˈsizijē/

Phil·os·o·nom·i·con /ˈfəˈläsə näm cän/

Words

multidisciplinary	*collaborator* *new tools* *complementary* *synergy*
interdisciplinary	*new field* *synthesis* *emergent*
transdisciplinary	*iconoclastic* *rebel* *transformative*

Jobs

	Positive	Negative
multidisciplinary	existing disciplines	existing disciplines
interdisciplinary	hot new field	no existing home
transdisciplinary	transformative	threatening

Case Study



Work

multidisciplinary	choosing partners
interdisciplinary	community
transdisciplinary	allies

Language

Theorem 1.1 (Solving Low-Degree Systems of Polynomial Equations). *Let p be a prime, and $q = p^d$ for $d \geq 1$. There is a randomized algorithm that, given an instance of SysPolyEqs(q) with m polynomial equations of degree at most k in n variables, decides the satisfiability of the system correctly with high probability. The running time of the algorithm is bounded by*

- $O^*(2^{0.8765n})$ *time*[1] *when* $q = k = 2$,
- $O^*(q^{\left(1-\frac{1}{5k}\right)n} \cdot n^{3k})$ *when* $p = 2$ *(but* $q > 2$ *or* $k > 2$*),*
- $O^*(q^{(1-(1/200k))n} \cdot n^{3qk})$ *when* $p > 2$ *and* $\log p < 4ek$,
- $O^*\left(q^n \cdot \left(\frac{\log q}{ekd}\right)^{-dn}\right)$ *when* $p > 2$ *and* $\log p \geq 4ek$

Lokshtanov *et al* SODA 2017

Dictionary

Translation dictionaries open up worlds.



Sean Hallgren
Theoretical CS
Penn State



Yaoyun Shi
Theoretical CS
Michigan

Hobbies

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arXiv.org

- Astrophysics
- Condensed Matter
- General Relativity and Quantum Cosmology
- HEP - Experiment
- HEP - Lattice
- HEP - Phenomenology
- HEP - Theory
- Mathematical Physics
- Nonlinear Sciences
- Nuclear Experiment
- Nuclear Theory
- More Physics
- Quantum Physics
- Mathematics
- Computer Science

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Topological proofs of contextuality in quantum mechanics

Scite! 13 PDF

Cihan Okay, Sam Roberts, Stephen D. Bartlett, Robert Raussendorf

Jan 10 2017 quant-ph arXiv:1701.01888v1

We provide a cohomological framework for contextuality of quantum mechanics that is suited to describing contextuality as a resource in measurement-based quantum computation. This framework applies to the parity proofs first discussed by Mermin, as well as a different type of contextuality proofs based on symmetry transformations. The topological arguments presented can be used in the state-dependent and the state-independent case.

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Scite! 9 PDF

Mathieu Lauriere, Dave Touchette

Jan 10 2017 quant-ph cs.CC cs.IT math.IT arXiv:1701.02062v1

In the context of two-party interactive quantum communication protocols, we study a recently defined notion of

Date Published

Tue 10 Jan 2017
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Overlapping qubits
Noon van der Silk 6 days ago

> This is a cool paper!

Freedom

“*Freedom's just another word for nothing left to lose*” - Janis Joplin



Whimsy



Create, Create, Create, Create



Imma Dog Says Thank You

